

Qualitative Salt Analysis Guide

Part I: Acidic Radicals (Anions)

A. Dilute H_2SO_4 Group

1. Carbonate (CO_3^{2-})

- Preliminary Test:** Add a few drops of dilute H_2SO_4 to a small quantity of the given salt in a test tube. Observation: Brisk effervescence occurs with the evolution of a colorless, odorless gas.
- Confirmatory Test (Lime Water Test):** Pass the evolved gas through freshly prepared lime water. Observation: The lime water turns milky due to the formation of insoluble calcium carbonate.



2. Sulphide (S^{2-})

- Preliminary Test:** Add dilute H_2SO_4 to the salt and warm slightly. Observation: A colorless gas with a distinct rotten egg smell is evolved.
- Confirmatory Test (Lead Acetate Test):** Bring a piece of filter paper moistened with lead acetate solution to the mouth of the test tube. Observation: The filter paper turns shining black.



3. Sulphite (SO_3^{2-})

- Preliminary Test:** Add dilute H_2SO_4 to the salt. Observation: A colorless gas with a suffocating smell of burning sulphur is evolved.
- Confirmatory Test (Potassium Dichromate Test):** Bring a filter paper dipped in acidified potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) solution near the mouth of the test tube. Observation: The filter paper turns from orange to green.



4. Nitrite (NO_2^-)

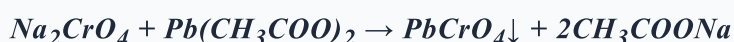
1. **Preliminary Test:** Add dilute H_2SO_4 to the salt. Observation: A reddish-brown gas with a pungent smell is evolved.
2. **Confirmatory Test (Starch Iodide Test):** Prepare an aqueous solution of the salt. Add a few drops of dilute H_2SO_4 , followed by potassium iodide (KI) solution and freshly prepared starch solution. Observation: A deep blue-black coloration appears.



B. Concentrated H_2SO_4 Group

1. Chloride (Cl^-)

1. **Preliminary Test:** Add concentrated H_2SO_4 to the salt and heat. Observation: A colorless, pungent-smelling gas evolves which gives dense white fumes when a glass rod dipped in NH_4OH is brought near it.
2. **Confirmatory Test (Chromyl Chloride Test):** Mix the salt with solid $\text{K}_2\text{Cr}_2\text{O}_7$ and add concentrated H_2SO_4 . Heat the mixture and pass the red vapors evolved into NaOH solution. Add acetic acid and lead acetate solution to the resulting yellow solution. Observation: A yellow precipitate of lead chromate forms.



2. Bromide (Br^-)

1. **Preliminary Test:** Add concentrated H_2SO_4 to the salt and heat. Observation: Reddish-brown vapors of bromine gas are evolved.
2. **Confirmatory Test (Silver Nitrate Test):** To the aqueous salt extract, add dilute HNO_3 followed by silver nitrate (AgNO_3) solution. Observation: A pale yellow precipitate forms which is partially soluble in ammonium hydroxide (NH_4OH) solution.

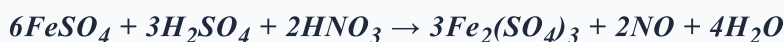


3. Nitrate (NO_3^-)

1. **Preliminary Test:** Add concentrated H_2SO_4 to the salt and heat. Add a few copper turnings.

Observation: The evolution of dense reddish-brown fumes (NO_2) intensifies.

2. **Confirmatory Test (Brown Ring Test):** Take the aqueous solution of the salt in a test tube. Add freshly prepared ferrous sulphate (FeSO_4) solution to it in equal volume. Carefully pour concentrated H_2SO_4 down the inner side of the inclined test tube. Observation: A dark brown ring forms at the junction of the two liquids.



4. Acetate (CH_3COO^-)

1. **Preliminary Test:** Take a small amount of salt on a watch glass, add a few drops of concentrated H_2SO_4 and rub it with a glass rod. Observation: A characteristic smell of vinegar is observed.
2. **Confirmatory Test (Ferric Chloride Test):** Add neutral ferric chloride (FeCl_3) solution to the aqueous salt extract. Observation: A deep red or blood-red coloration is produced.



C. Independent Group

1. Sulphate (SO_4^{2-})

1. **Test:** To the aqueous solution of the salt, add dilute HCl and boil. Then, add barium chloride (BaCl_2) solution.
2. **Observation:** A thick white precipitate forms which remains insoluble even upon adding concentrated HCl .



2. Phosphate (PO_4^{3-})

1. **Test:** To the aqueous salt solution, add concentrated HNO_3 and boil. Add ammonium molybdate solution to the hot mixture and heat again.
2. **Observation:** A canary yellow precipitate of ammonium phosphomolybdate is formed.



Part II: Basic Radicals (Cations)

Group o

1. Ammonium (NH_4^+)

- Preliminary Test:** Heat a small quantity of the original salt with sodium hydroxide (NaOH) solution. Observation: A colorless gas with a pungent, choking smell of ammonia is evolved.
- Confirmatory Test (Nessler's Reagent Test):** Pass the evolved gas through Nessler's reagent (or add the reagent directly to the salt solution). Observation: A brown precipitate or deep brown coloration is obtained.



Group I

1. Lead (Pb^{2+})

- Preliminary Test:** Prepare the Original Solution (O.S.) of the salt in water. Add dilute HCl. Observation: A white precipitate of lead chloride forms, which dissolves on heating and reappears on cooling.
- Confirmatory Test (Potassium Chromate Test):** Dissolve the white precipitate in hot water. Add potassium chromate (K_2CrO_4) solution. Observation: A brilliant yellow precipitate of lead chromate forms.



Group II

1. Copper (Cu^{2+})

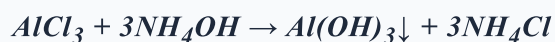
- Preliminary Test:** Pass H_2S gas through the acidified original solution. Observation: A black precipitate of CuS forms. Dissolve this precipitate in 50% nitric acid and boil off the H_2S gas.
- Confirmatory Test (Ammonia Test):** To the above clear solution, add ammonium hydroxide (NH_4OH) solution dropwise, then in excess. Observation: A pale blue precipitate forms initially, which dissolves in excess reagent to yield a deep blue (ink blue) colored solution.



Group III

1. Aluminium (Al^{3+})

- Preliminary Test:** Boil off H_2S from the Group II filtrate. Add a few drops of concentrated HNO_3 and boil. Cool, add solid NH_4Cl , and then add NH_4OH in excess. Observation: A gelatinous white precipitate forms. Dissolve it in dilute HCl .
- Confirmatory Test (Lake Test):** To the dissolved solution, add 2 drops of blue litmus solution, followed by NH_4OH dropwise until the solution is alkaline. Observation: A blue mass (precipitate) floats suspended in a colorless solution.



2. Ferric (Fe^{3+})

- Preliminary Test:** Follow the same precipitation steps as for Aluminium. Observation: A reddish-brown precipitate forms. Dissolve it in dilute HCl .
- Confirmatory Test (Prussian Blue Test):** To the dissolved solution, add potassium ferrocyanide ($\text{K}_4[\text{Fe}(\text{CN})_6]$) solution. Observation: A deep blue (Prussian blue) precipitate or coloration is formed.



Group IV

1. Zinc (Zn^{2+})

- Preliminary Test:** Pass H_2S gas through the ammoniacal solution of Group III. Observation: A dirty white or greyish-white precipitate forms. Dissolve this in dilute HCl and boil off H_2S .
- Confirmatory Test (NaOH Test):** Add sodium hydroxide (NaOH) solution dropwise, and then in excess, to the clear solution. Observation: A white precipitate forms initially, which dissolves completely in excess NaOH to form a clear soluble solution of sodium zincate.

